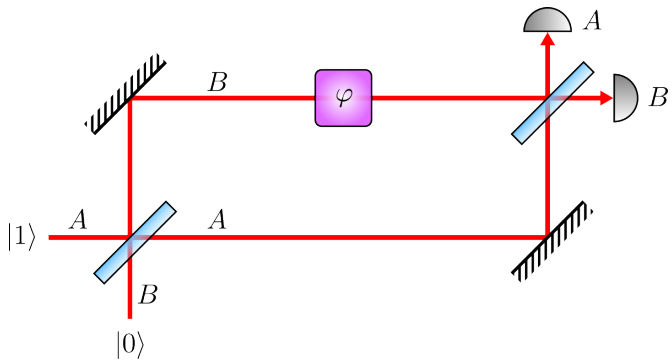
Note that, since the estimator is unbiased, the variance also equals the mean-squared error $\mathrm{MSE}(\tilde{\varphi}) = \Delta^2\tilde{\varphi}$. The Fisher information thus sets a lower bound on the best possible precision we can hope to obtain from a given measurement. An estimator which achives equality in the Cramér-Rao bound is said to be *efficient*.

Because the Fisher information is additive for independent measurements (see Prob. 4.2), if a parameter is estimated by repeating the same experiment N times, with each repetition corresponding to X, then

$$\langle\Delta^2\tilde{\varphi}\rangle \geq \frac{1}{N\mathcal{F}_\varphi(X)}. \tag{4.19}$$

The variance of the best estimate thus improves as $\frac{1}{N}$.

Example 4.3: Phase estimation with single photons



Consider the task of estimating the phase difference between two optical paths. This can be done using a single photon in a setup as shown (known as a Mach-Zender interferometer). The photon is first split on a balanced beam splitter into a superposition of two paths. One path (B) undergoes a phase shift φ while the other (A) does not. The paths are the recombined on a second beam splitter and the outputs are measured with single-photon detectors. What precision can be achieved for estimating φ with this strategy?

Assuming no loss, the state transforms as

$$|1,0\rangle_{AB} \underset{\text{first BS}}{\longrightarrow} \frac{1}{\sqrt{2}}\left(|1,0\rangle_{AB} + |0,1\rangle_{AB}\right) \tag{4.20}$$

$$\underset{\text{phase shift}}{\longrightarrow} \frac{1}{\sqrt{2}}\left(|1,0\rangle_{AB} + e^{i\varphi}|0,1\rangle_{AB}\right) \tag{4.21}$$

$$\underset{\text{second BS}}{\longrightarrow} \frac{1+e^{i\varphi}}{2}|1,0\rangle_{AB} + \frac{1-e^{i\varphi}}{2}|0,1\rangle_{AB} \tag{4.22}$$

$$= e^{i\frac{\varphi}{2}}\left[\cos\left(\frac{\varphi}{2}\right)|1,0\rangle_{AB} + i\sin\left(\frac{\varphi}{2}\right)|0,1\rangle_{AB}\right]. \tag{4.23}$$

Assuming perfect detectors, there are two possible outcomes: a single click at A or B (denoted 0 and 1, respectively) with probabilities

$$p_\varphi(0) = \cos^2(\frac{\varphi}{2}) \quad \text{and} \quad p_\varphi(1) = \sin^2(\frac{\varphi}{2}). \tag{4.24}$$

From (4.1)-(4.2), the Fisher information is

$$\mathcal{F}_\varphi = \frac{[2\cos\left(\frac{\varphi}{2}\right)\sin\left(\frac{\varphi}{2}\right)\frac{1}{2}]^2}{\cos^2(\frac{\varphi}{2})} + \frac{[-2\cos\left(\frac{\varphi}{2}\right)\sin\left(\frac{\varphi}{2}\right)\frac{1}{2}]^2}{\sin^2(\frac{\varphi}{2})} = 1. \tag{4.25}$$

The Cramér-Rao bound then stipulates that for N independent repetitions of this experiment (i.e. using N single photons independently) and any unbiased estimator $\langle\Delta^2\tilde{\varphi}\rangle \geq 1/N$. This can be improved by instead using N photons to prepare multi-photon entangled states, as discussed in Sec. 4.2 and Chap. 5.

Proof for unbiased estimators

The Cramér-Rao bound follows from the Cauchy-Schwartz inequality for random variables. The Cauchy-Schwartz inequality states that for any pair of random variables A, B

$$\mathrm{Cov}(A,B) \leq \langle\Delta^2 A\rangle\langle\Delta^2 B\rangle, \tag{4.26}$$

where $\langle\Delta^2 A\rangle = \langle(A-\langle A\rangle)^2\rangle$ denotes the variance as usual (similarly for B) and $\mathrm{Cov}(A,B) = \langle(A-\langle A\rangle)(B-\langle B\rangle)\rangle$ the covariance.

Take the random variables $A = \tilde{\varphi}(X)$ and $B = \frac{\partial}{\partial\varphi}\log(p_\varphi(X))$. Then by definition $\mathrm{Var}(A) = \Delta^2\tilde{\varphi}$ and $\mathrm{Var}(B) = \mathcal{F}_\varphi(X)$. Because $\tilde{\varphi}$ is unbiased, $\langle A\rangle = \varphi$, and because the mean score is zero, $\langle B\rangle = 0$. The covariance is therefore

$$\begin{aligned}
\mathrm{Cov}(A,B) &= \langle(\tilde{\varphi}(X)-\varphi)\frac{\partial}{\partial\varphi}\log p_\varphi(X)\rangle && (4.27)\\
&= \langle\tilde{\varphi}(X)\frac{\partial}{\partial\varphi}\log p_\varphi(X)\rangle && (4.28)\\
&= \sum_x \tilde{\varphi}(x)\frac{\partial}{\partial\varphi}p_\varphi(x) && (4.29)\\
&= \frac{\partial}{\partial\varphi}\sum_x \tilde{\varphi}(x)p_\varphi(x) && (4.30)\\
&= \frac{\partial}{\partial\varphi}\varphi = 1. && (4.31)
\end{aligned}$$

The Cauchy-Schwartz inequality then implies

$$\langle\Delta^2\tilde{\varphi}\rangle\,\mathcal{F}_\varphi(X) \geq 1, \tag{4.32}$$

and the Cramér-Rao bound follows.

Biased estimators

The Cramér-Rao bound can be generalised to estimators with nonzero bias. In this case, the mean-squared error and the variance do not coincide, c.f. Eq. (4.12). It is trivial to obtain a small variance. For example, an estimator which just outputs a constant independent of φ will have zero variance but

Chapter 5

Sensing with light

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Exercises

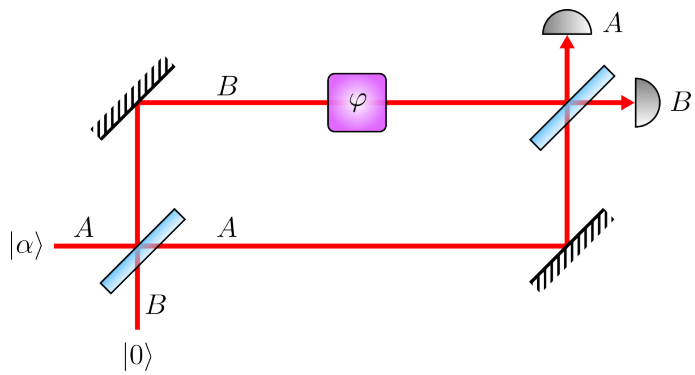

Figure 5.1: Mach-Zender interferometer with a coherent state entering one input port. Both beam splitters are balanced.

Problem 5.1: Coherent and squeezed-state interferometry

Consider a Mach-Zender interferometer as shown in Fig. 5.1, where a coherent state $|\alpha\rangle$ enters one input while vacuum is injected in the other input port. A small phase shift φ is applied in one of the interferometer arms. We take the beam splitters to be described by the unitary $\hat{U}_{BS}(\frac{1}{2})$ in Eqs. (3.134)-(3.135) (with the roles of A and B reversed at the second beam splitter) and the unitary encoding the phase shift to be $\hat{R}(-\varphi)$ (c.f. Sec. 3.1.4).

(a) Show that the probability of detecting n photons at output A as a function of φ, assuming an ideal photon counter and no losses, is given by

$$p_\varphi(n) = e^{-|\alpha|^2 \sin^2(\frac{\varphi}{2})} \frac{|\alpha|^{2n} \sin^{2n}(\frac{\varphi}{2})}{n!}. \tag{5.1}$$

(b) Find the Fisher information corresponding to the photon counting measurement. Show that the estimate variance for φ obeys the shot noise limit.

$$\langle \Delta^2 \tilde{\varphi} \rangle \geq \frac{1}{|\alpha|^2}. \tag{5.2}$$

(c) Consider now that the vacuum input in B is replaced by some arbitrary state $|\psi\rangle$. Assuming that α is real, show that the expected number of photons at output A can be written

$$\langle \hat{n}_A \rangle = \alpha^2 \sin^2(\frac{\varphi}{2}) + \langle \hat{n}_B \rangle_\psi \cos^2(\frac{\varphi}{2}) - \sqrt{2}\alpha \sin\Big(\frac{\varphi}{2}\Big) \cos\Big(\frac{\varphi}{2}\Big) \langle \hat{p}_B \rangle_\psi, \tag{5.3}$$

where $\langle \cdot \rangle_\psi = \langle \psi | \cdot | \psi \rangle$ denotes the expectation value in state $|\psi\rangle$ and $\hat{n}_B$ and $\hat{p}_B$ are the number and phase quadrature operators of mode B, respectively.

(d) Assume that $|\psi\rangle$ has support only on even photon-number (Fock) states, as is the case e.g. for vacuum and squeezed-vacuum states. Show that then

$$\begin{aligned}\langle \hat{n}_A^2\rangle =& \sin^4(\frac{\varphi}{2})(\alpha^2+\alpha^4)+\cos^4(\frac{\varphi}{2})\langle \hat{n}_B^2\rangle_\psi \\ &+2\sin^2(\frac{\varphi}{2})\cos^2(\frac{\varphi}{2})\alpha^2\langle \hat{n}_B\rangle_\psi \\ &+\sin^2(\frac{\varphi}{2})\cos^2(\frac{\varphi}{2})\left[2\alpha^2\langle \hat{p}_B^2\rangle_\psi+\langle \hat{n}_B\rangle_\psi\right].\end{aligned} \tag{5.4}$$

(e) Show that for vacuum input, i.e. $|\psi\rangle = |0\rangle$, we have the mean and variance at output A

$$\langle \hat{n}_A\rangle = \alpha^2\sin^2(\frac{\varphi}{2}), \tag{5.5}$$

$$\langle \Delta^2\hat{n}_A\rangle = \alpha^2\sin^2(\frac{\varphi}{2}). \tag{5.6}$$

(f) Let now $|\psi\rangle$ be a phase-squeezed vacuum state (i.e. squeezed in $\hat{p}$) with squeezing parameter r. Show that for sufficiently small phase φ and large α we have

$$\langle \hat{n}_A\rangle \approx \frac{1}{4}\alpha^2\varphi^2, \tag{5.7}$$

$$\langle \Delta^2\hat{n}_A\rangle \approx \frac{1}{2}\alpha^2\langle \hat{p}_B^2\rangle_\psi\varphi^2. \tag{5.8}$$

Hence, estimating φ from the photocurrent at output A gives

$$\langle \Delta^2\tilde{\varphi}\rangle = \frac{\langle \Delta^2\hat{n}_{\mathrm{A_{out}}}\rangle}{\left|\frac{\partial\langle \hat{n}_A\rangle}{\partial\varphi}\right|^2} \approx \frac{2\langle \hat{p}_B^2\rangle_\psi}{\alpha^2} = \frac{e^{-2r}}{\alpha^2}. \tag{5.9}$$

That is, the estimate variance is suppressed below shot noise by the squeezing factor.

Problem 5.2: QFI for NOON states

Show that the quantum Fisher information of the phase-dependent NOON state $\frac{1}{\sqrt{2}}(|N,0\rangle + e^{i\varphi N}|0,N\rangle)$ with respect to φ equals $\mathcal{Q}_\varphi = N^2$.

Problem 5.3: Photon loss in interferometry

We have seen that, in the absence of imperfections, interferometric phase estimation with N-photon NOON-states

$$|\psi_{NOON}\rangle = \frac{1}{\sqrt{2}}(|N,0\rangle + |0,N\rangle) \tag{5.10}$$

yields an estimate variance, for ν repetitions, scaling as

$$\langle \Delta^2\tilde{\varphi}\rangle_{NOON} \geq \frac{1}{\nu N^2}. \tag{5.11}$$

Using coherent states with mean photon number $N = |\alpha|^2$ (in a Mach-Zender

interferometer), on the other hand, yields

$$\langle \Delta^2 \tilde{\varphi} \rangle_\alpha \geq \frac{1}{\nu N}. \tag{5.12}$$

Ideally, NOON states thus achieve a quadratic improvement with the (mean) photon number. The sensitivities to loss, however, are very different for the two approaches.

(a) Find the state resulting from applying symmetric loss to the two modes of the NOON state (5.10) Argue that the loss of any non-zero number of photons will render the state useless for sensing the phase. *Hint: loss can be modelled by each mode passing through a beam splitter with transmittivity η – the other input of the beam splitter is vacuum and the other output is traced out.*

(b) Show that the estimate variance for the NOON state is bounded by

$$\langle \Delta^2 \tilde{\varphi} \rangle_{NOON} \geq \frac{1}{\nu \eta^N N^2}, \tag{5.13}$$

for efficiency η.

(c) Similarly, find the scaling of $\langle \Delta^2 \tilde{\varphi} \rangle_\alpha$ with the efficiency η, assuming symmetric losses in the two interferometer arms.

(d) When is it advantageous to use one or the other approach?